

- [Skip to main navigation](#)
- [Skip to search](#)
- [Skip to main content](#)

[Coventry University Home](#) [Coventry University Logo](#)

- [Help & FAQ](#)
- [Home](#)
- [Organisational Units](#)
- [Profiles](#)
- [Research output](#)
- [Projects](#)
- [Coventry awarded Student theses](#)
- [Activities](#)
- [Prizes](#)
- [Datasets](#)
- [Press/Media](#)
- [Equipment](#)

Search by expertise, name or affiliation

“Maze Out”: A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders

Maria Guala, Aida Bikic, [Kim Bul](#), David Clinton, Anna Mejdal, Angelina Isabella Mellentin, Helene Nygaard Nielsen, Elsebeth Stenager, Anette Søgaard Nielsen

- [Centre for Intelligent Healthcare](#)
- [University of Southern Denmark](#)
- [Karolinska Institutet](#)

Research output: Contribution to journal › Article › peer-review

12 Downloads (Pure)

- [Overview](#)
- [Fingerprint](#)

Abstract

Background: Eating Disorders (ED) are severe and costly mental health disorders. The effects of existing treatment approaches are limited and there is a need to develop novel interventions, including digital strategies that can increase engagement and effectiveness. Maze Out is a new serious game coproduced by patients and ED therapists, which allows patients to “play” with the reality of an ED and reflect on associated challenges.

Objectives: The present study has two main objectives: 1) to evaluate the effectiveness of adding Maze Out to treatment as usual (TAU) in a randomised controlled trial (RCT); and 2) to examine in depth the potential of Maze Out by examining how it is perceived and used in the context of an RCT.

Methods: Participants will be recruited from mental health care services, endocrinology departments or Community Centres offering treatment for ED. Patients suffering from ED (N=94) will be randomised to either TAU or TAU plus Maze Out. Primary outcome will be measured in terms of changes in self-efficacy, measured by a 5-item self-efficacy questionnaire (5-item SE_ED). Secondary outcome measures will include feelings of ineffectiveness and self-image, as measured by Brief INSPIRE-O and Structural Analysis of Social Behaviour Intrex Questionnaire (SASB). Data will be collected at baseline (enrolment in the study), and subsequently 8 and 15 weeks after inclusion. Experiences of playing Maze Out will be examined in a sub-sample (n=30) of participants, utilising both quantitative user analytics and qualitative interview data of patients, interview data of significant others, and healthcare professionals to explore the possible impact of Maze Out on disorder insight, communication patterns between patients and therapists and understanding of their disorder.

Discussion: To our knowledge Maze Out is the first serious game coproduced by patients and therapists. It is a novel and theoretically grounded intervention that may significantly contribute to the healing process of ED. If found effective, the potential for wide-spread impact and scalability is considerable.

Trial registration: ClinicalTrials.gov NCT05621018

Original language English

Article number 35

Number of pages 10

Journal [Journal of Eating Disorders](#)

Volume 12

DOIs • <https://doi.org/10.1186/s40337-024-00985-2>

Publication status Published - 1 Mar 2024

Bibliographical note

This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>. The Creative Commons Public Domain Dedication waiver (<http://creativecommons.org/publicdomain/zero/1.0/>) applies to the data made available in this article, unless otherwise stated in a credit line to the data.

Funder

Open access funding provided by University of Southern Denmark The study is funded by TrygFonden, the Psychiatric Research Foundation, Region of Southern Denmark and, Helsefonden. The funding organizations are independent and have no involvement in the design, data collection or analysis of the current trial.

Keywords

- eating disorders
- serious games
- self-efficacy
- co-production
- randomised controlled trial

UN SDGs

This output contributes to the following UN [Sustainable Development Goals \(SDGs\)](#)

- SDG 3 - Good Health and Well-being

Access to Document

- [10.1186/s40337-024-00985-2](#)Licence: [CC BY](#)
- [Bul2024VoR](#)Final published version, 1.39 MBLicence: [CC BY](#)
- [Guala_et_al-2024-Journal_of_Eating_Disorders](#)Final published version, 1.02 MBLicence: [CC BY](#)

Other files and links

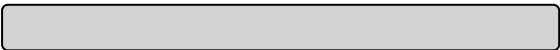
- [Link to publication in Scopus](#)

Fingerprint

Dive into the research topics of "'Maze Out": A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders'. Together they form a unique fingerprint.

- 

Treatment of Eating Disorders
Medicine and Dentistry
100%

- 

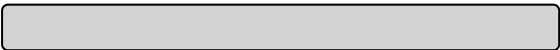
Randomized Controlled Trial
Medicine and Dentistry
100%

- 

Eating Disorder
Medicine and Dentistry
100%

- 

Eating Disorders
Psychology
100%

- 

Patient
Medicine and Dentistry
85%

- 

Therapeutic Procedure
Medicine and Dentistry
71%

- 

Treatment
Psychology
71%

- 

Self Concept
Medicine and Dentistry
42%

[View full fingerprint](#)

Cite this

- APA
- Standard
- Harvard
- Vancouver
- Author
- BIBTEX
- RIS

Guala, M., Bikic, A., [Bul, K.](#), Clinton, D., Mejdal, A., Mellentin, A. I., Nygaard Nielsen, H., Stenager, E., & Nielsen, A. S. (2024). [“Maze Out”: A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders.](#) **Journal of Eating Disorders**, 12, Article 35 . <https://doi.org/10.1186/s40337-024-00985-2>
[“Maze Out”: A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders.](#) / Guala, Maria ; Bikic, Aida; [Bul, Kim](#) et al.
 In: [Journal of Eating Disorders](#), Vol. 12, 35 , 01.03.2024.

Research output: Contribution to journal › Article › peer-review

Guala, M, Bikic, A, [Bul, K](#), Clinton, D, Mejdal, A, Mellentin, AI, Nygaard Nielsen, H, Stenager, E & Nielsen, AS 2024, '[“Maze Out”: A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the](#)

treatment of eating disorders', [Journal of Eating Disorders](https://doi.org/10.1186/s40337-024-00985-2), vol. 12, 35 .
<https://doi.org/10.1186/s40337-024-00985-2>

Guala M, Bikic A, [Bul K](#), Clinton D, Mejdal A, Mellentin AI et al. [“Maze Out”: A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders.](#) [Journal of Eating Disorders](#). 2024 Mar 1;12:35 . doi: 10.1186/s40337-024-00985-2

Guala, Maria ; Bikic, Aida ; [Bul, Kim](#) et al. / [“Maze Out” : A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders.](#) In: [Journal of Eating Disorders](#). 2024 ; Vol. 12.

@article{5b908b0090d042908a2a2f226fc6b45a,

title = "“Maze Out”: A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders",

abstract = "Background: Eating Disorders (ED) are severe and costly mental health disorders. The effects of existing treatment approaches are limited and there is a need to develop novel interventions, including digital strategies that can increase engagement and effectiveness. Maze Out is a new serious game coproduced by patients and ED therapists, which allows patients to “play” with the reality of an ED and reflect on associated challenges.Objectives: The present study has two main objectives: 1) to evaluate the effectiveness of adding Maze Out to treatment as usual (TAU) in a randomised controlled trial (RCT); and 2) to examine in depth the potential of Maze Out by examining how it is perceived and used in the context of an RCT.Methods: Participants will be recruited from mental health care services, endocrinology departments or Community Centres offering treatment for ED. Patients suffering from ED (N=94) will be randomised to either TAU or TAU plus Maze Out. Primary outcome will be measured in terms of changes in self-efficacy, measured by a 5-item self-efficacy questionnaire (5-item SE_ED). Secondary outcome measures will include feelings of ineffectiveness and self-image, as measured by Brief INSPIRE-O and Structural Analysis of Social Behaviour Intrex Questionnaire (SASB). Data will be collected at baseline (enrolment in the study), and subsequently 8 and 15 weeks after inclusion. Experiences of playing Maze Out will be examined in a sub-sample (n=30) of participants, utilising both quantitative user analytics and qualitative interview data of patients, interview data of significant others, and healthcare professionals to explore the possible impact of Maze Out on disorder insight, communication patterns between patients and therapists and understanding of their disorder. Discussion: To our knowledge Maze Out is the first serious game coproduced by patients and therapists. It is a novel and theoretically grounded intervention that may significantly contribute to the healing process of ED. If found effective, the potential for wide-spread impact and scalability is considerable. Trial registration: ClinicalTrials.gov NCT05621018 ",

keywords = "eating disorders, serious games, self-efficacy, co-production, randomised controlled trial",

author = "Maria Guala and Aida Bikic and Kim Bul and David Clinton and Anna Mejdal and Mellentin, {Angelina Isabella} and {Nygaard Nielsen}, Helene and Elsebeth Stenager and Nielsen, {Anette S{\o}gaard}",

note = "This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>. The Creative Commons Public Domain Dedication waiver (<http://creativecommons.org/publicdomain/zero/1.0/>) applies to the data made available in this article, unless otherwise stated in a credit line to the data. ",

year = "2024",
month = mar,
day = "1",
doi = "10.1186/s40337-024-00985-2",
language = "English",
volume = "12",
journal = "Journal of Eating Disorders",
issn = "2050-2974",
publisher = "BioMed Central Ltd.",

}

TY - JOUR

T1 - "Maze Out"

T2 - A study protocol for a randomised controlled trial using a mix methods approach exploring the potential and examining the effectiveness of a serious game in the treatment of eating disorders

AU - Guala, Maria

AU - Bikic, Aida

AU - Bul, Kim

AU - Clinton, David

AU - Mejdal, Anna

AU - Mellentin, Angelina Isabella

AU - Nygaard Nielsen, Helene

AU - Stenager, Elsebeth

AU - Nielsen, Anette Sogaard

N1 - This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>. The Creative Commons Public Domain Dedication waiver (<http://creativecommons.org/publicdomain/zero/1.0/>) applies to the data made available in this article, unless otherwise stated in a credit line to the data.

PY - 2024/3/1

Y1 - 2024/3/1

N2 - Background: Eating Disorders (ED) are severe and costly mental health disorders. The effects of existing treatment approaches are limited and there is a need to develop novel interventions, including digital strategies that can increase engagement and effectiveness. Maze Out is a new serious game coproduced by patients and ED therapists, which allows patients to “play” with the reality of an ED and reflect on associated challenges. Objectives: The present study has two main objectives: 1) to evaluate the effectiveness of adding Maze Out to treatment as usual (TAU) in a randomised controlled trial (RCT); and 2) to examine in depth the potential of Maze Out by examining how it is perceived and used in the context of an RCT. Methods: Participants will be recruited from mental health care services, endocrinology departments or Community Centres offering treatment for ED. Patients suffering from ED (N=94) will be randomised to either TAU or TAU plus Maze Out. Primary outcome will be measured in terms of changes in self-efficacy, measured by a 5-item self-efficacy questionnaire (5-item SE_ED). Secondary outcome measures will include feelings of ineffectiveness and self-image, as measured by Brief INSPIRE-O and Structural Analysis of Social Behaviour Intrex Questionnaire (SASB). Data will be collected at baseline (enrolment in the study), and subsequently 8 and 15 weeks after inclusion. Experiences of playing Maze Out will be examined in a sub-sample (n=30) of participants, utilising both quantitative user analytics and qualitative interview data of patients, interview data of significant others, and healthcare professionals to explore the possible impact of Maze Out on disorder insight, communication patterns between patients and therapists and understanding of their disorder. Discussion: To our knowledge Maze Out is the first serious game coproduced by patients and therapists. It is a novel and theoretically grounded intervention that may significantly contribute to the healing process of ED. If found effective, the potential for wide-spread impact and scalability is considerable. Trial registration: ClinicalTrials.gov NCT05621018

AB - Background: Eating Disorders (ED) are severe and costly mental health disorders. The effects of existing treatment approaches are limited and there is a need to develop novel interventions, including digital strategies that can increase engagement and effectiveness. Maze Out is a new serious game coproduced by patients and ED therapists, which allows patients to “play” with the reality of an ED and reflect on associated challenges.**Objectives:** The present study has two main objectives: 1) to evaluate the effectiveness of adding Maze Out to treatment as usual (TAU) in a randomised controlled trial (RCT); and 2) to examine in depth the potential of Maze Out by examining how it is perceived and used in the context of an RCT.**Methods:** Participants will be recruited from mental health care services, endocrinology departments or Community Centres offering treatment for ED. Patients suffering from ED (N=94) will be randomised to either TAU or TAU plus Maze Out. Primary outcome will be measured in terms of changes in self-efficacy, measured by a 5-item self-efficacy questionnaire (5-item SE_ED). Secondary outcome measures will include feelings of ineffectiveness and self-image, as measured by Brief INSPIRE-O and Structural Analysis of Social Behaviour Intrex Questionnaire (SASB). Data will be collected at baseline (enrolment in the study), and subsequently 8 and 15 weeks after inclusion. Experiences of playing Maze Out will be examined in a sub-sample (n=30) of participants, utilising both quantitative user analytics and qualitative interview data of patients, interview data of significant others, and healthcare professionals to explore the possible impact of Maze Out on disorder insight, communication patterns between patients and therapists and understanding of their disorder. **Discussion:** To our knowledge Maze Out is the first serious game coproduced by patients and therapists. It is a novel and theoretically grounded intervention that may significantly contribute to the healing process of ED. If found effective, the potential for wide-spread impact and scalability is considerable. Trial registration: ClinicalTrials.gov NCT05621018

KW - eating disorders

KW - serious games

KW - self-efficacy

KW - co-production

KW - randomised controlled trial

UR - <http://www.scopus.com/inward/record.url?scp=85186616320&partnerID=8YFLogxK>

U2 - 10.1186/s40337-024-00985-2

DO - 10.1186/s40337-024-00985-2

M3 - Article

SN - 2050-2974

VL - 12

JO - Journal of Eating Disorders

JF - Journal of Eating Disorders

M1 - 35

ER -



Powered by [Pure](#), [Scopus](#) & [Elsevier Fingerprint Engine™](#)

All content on this site: Copyright © 2024 [Elsevier B.V.](#) or its licensors and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the Creative Commons licensing terms apply

We use cookies to help provide and enhance our service and tailor content. By continuing you agree to the [use of cookies](#)



[Log in to Pure](#)

[Coventry University data protection policy](#)

[About web accessibility](#)

[Report vulnerability](#)

[Contact us](#)